Probability Limit Theorems

MATH-SHU 350

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1 Measure Theory and Probability Spaces

1.1 Measure Theory

1.1.1 Basic Concepts

Definition 1.1. Let E be a set, and let $\mathscr{A}$ be a set of subsets of E. We say that

$$\text{if} \quad \varnothing \in \mathscr{A}, \quad A^C \in \mathscr{A},\ A \cup B \in \mathscr{A}, \quad \forall A, B \in \mathscr{A}, \quad \text{then } \mathscr{A} \text{ is an } \textbf{algebra}, \tag{1}$$

$$\text{if} \quad \varnothing \in \mathscr{A}, \quad A \cap B \in \mathscr{A}, \quad \forall A, B \in \mathscr{A}, \quad \text{then } \mathscr{A} \text{ is a } \pi\textbf{-system}, \tag{2}$$

$$\text{if} \quad \varnothing \in \mathscr{A}, \quad A^C \in \mathscr{A},\ \bigcup_{n\in\mathbb{N}} A_n \in \mathscr{A}, \quad \forall A, A_n \in \mathscr{A}, \quad \text{then } \mathscr{A} \text{ is a } \sigma\textbf{-algebra}. \tag{3}$$

Definition 1.2. A pair $(E, \mathscr{E})$, where E is a set and $\mathscr{E}$ is a σ-algebra on E, is called a **measurable space**. In this case, each $A \in \mathscr{E}$ is called an $\mathscr{E}$**-measurable set**.

Definition 1.3. The intersection of all σ-algebras containing $\mathscr{C}$ (there is at least one) is called the σ**-algebra generated by** $\mathscr{C}$, denoted by $\sigma(\mathscr{C})$.

Remark 1.4. Note that $\mathscr{P}(E)$ (the set of all subsets of E) is a σ-algebra, and the intersection of any collection of σ-algebras is a σ-algebra. $\sigma(\mathscr{C})$ is the smallest σ-algebra containing $\mathscr{C}$.

Remark 1.5. It is commonly impossible to write down the typical element of a σ-algebra $\mathscr{E}$, so we try to find π-systems with simpler elements that generate our σ-algebra and work with them instead.

Definition 1.6. A **topological space** E with **topology** $\mathscr{T}$ is a space endowed with a set of open subsets. The **Borel** σ**-algebra** is then defined as $\mathscr{B}(E) = \sigma(\mathscr{T})$, which is the σ-algebra generated by the family of all open subsets of E. We use the standard abbreviation $\mathscr{B} := \mathscr{B}(\mathbb{R})$.

Example 1.7. Let $\pi(\mathbb{R})$ be the π-system $\pi(\mathbb{R}) := \{(-\infty, x];\ x \in \mathbb{R}\}$. Check that $\sigma(\pi(\mathbb{R})) = \mathscr{B}(\mathbb{R})$, *i.e.*, $\sigma(\pi(\mathbb{R}))$ is the Borel σ-algebra of $\mathbb{R}$.

Definition 1.8. Let E be a set and $\mathscr{A}$ be an algebra on E. A **set function** is any function $\mu : \mathscr{A} \to [0, \infty]$ with $\mu(\varnothing) = 0$.

Definition 1.9. Let E be a set, $\mathscr{A}$ be an algebra on E, and μ be a set function on $\mathscr{A}$. We say that

$$\begin{aligned}
&\mu \text{ is } \textbf{increasing} && \text{if} && \mu(A) \leq \mu(B), && \forall A, B \in \mathscr{A} \text{ with } A \subseteq B, && (4)\\
&\mu \text{ is } \textbf{additive} && \text{if} && \mu(A \cup B) = \mu(A) + \mu(B), && \forall A, B \in \mathscr{A} \text{ disjoint}, && (5)\\
&\mu \text{ is } \sigma\textbf{-additive} && \text{if} && \mu\left(\bigcup_{n\in\mathbb{N}} A_n\right) = \sum_{n\in\mathbb{N}} \mu(A_n), && \forall \{A_n\}_{n\in\mathbb{N}} \subseteq \mathscr{A} \text{ disjoint, with } \bigcup_{n\in\mathbb{N}} A_n \in \mathscr{A}. && (6)
\end{aligned}$$

Remark 1.10. σ-additivity is also called countable additivity. Let $(E, \mathscr{E})$ be a measurable space, then if $\mu : \mathscr{E} \to [0, \infty]$ is countable additive, it is called a **measure** and the triple $(E, \mathscr{E}, \mu)$ is called a **measure space**. In particular if $\mu(E) = 1$, then μ is called a **probability measure** and $(E, \mathscr{E}, \mu)$ is called a **probability space**. The notation $(\Omega, \mathscr{F}, \mathbb{P})$ is often used instead for probability spaces.

Remark 1.11. Given a measure space $(E, \mathscr{E}, \mu)$, μ is said to be **finite** if $\mu(E) < \infty$, and **σ-finite** if there exists $\{E_n\}_{n\in\mathbb{N}} \subseteq \mathscr{E}$, such that $\mu(E_n) < \infty$ and $\bigcup_{n\in\mathbb{N}} E_n = E$. A set $F \in \mathscr{E}$ is called **μ-null** if $\mu(F) = 0$.

Remark 1.12. A statement ϕ about points $s \in E$ holds **almost everywhere** (*a.e.*) if $F = \{s \in E;\ \phi(s) \text{ is false}\} \in \mathscr{E}$ and F is μ-null, *i.e.*, $\mu(F) = 0$. For a probability space, almost everywhere is often called **almost surely** (*a.s.*).

Example 1.13 (Infinitely often)**.** A_n occurs infinitely often if $\forall p \in \mathbb{N}$, $\exists n \geq p$, such that A_n holds. Now look at the event that an element $\omega \in$ infinitely many A_n's. This is equivalent to $\forall p \in \mathbb{N}$, $\exists n \geq p$, such that $\omega \in A_n$. Then $\forall p \in \mathbb{N}$, $\omega \in \bigcup_{n\geq p} A_n$, *i.e.*, $\omega \in \bigcap_{p\in\mathbb{N}} \bigcup_{n\geq p} A_n$. Therefore, we can see that

$$\{\omega;\ \omega \in A_n \text{ infinitely often}\} = \bigcap_{p\in\mathbb{N}} \bigcup_{n\geq p} A_n. \tag{7}$$

Note that we can interpret unions and intersections as

$$\bigcup_{n\in\mathbb{N}} A_n = \{A_n \text{ occurs at least once}\}, \qquad \bigcap_{n\in\mathbb{N}} A_n = \{\text{all } A_n\text{'s occur}\}. \tag{8}$$

Example 1.14 (A set we cannot measure on $\mathbb{R}$)**.** Take the equivalence class $x \sim y \iff y - x \in \mathbb{Q}$. The axiom of choice enables us to define a set C, such that C as exactly one element in each equivalent class, and we assume that $C \subseteq [0, 1]$ (applicable by translation). Assume $\mu(C) = 0$ and we define for some $r \in \mathbb{R}$ that

$$C_r = C + r = \{x \in \mathbb{R};\ x \in \mathbb{R},\ a \in C\}. \tag{9}$$

We can CHECK THAT C_r, $r \in \mathbb{Q}$ are disjoint. Indeed, if we take $x \in C_r \cap C_s$, then we can write $x = c + r = c' + s$ for some $c, c' \in C$. This would imply that $c' = c + r - s$, meaning that $r = s$ since there is can be only one element in each equivalent class. This leads to a contradiction, so that $C_r \cap C_s = \varnothing$, $\forall r, s \in \mathbb{Q}$. Then we can see that

$$\mu(\mathbb{R}) = \mu\left(\bigcup_{r\in\mathbb{Q}} C_r\right) = \sum_{r\in\mathbb{Q}} \mu(C_r) = \underbrace{\sum_{r\in\mathbb{Q}} \mu(C)}_{\text{translation invariance}} = \sum_{r\in\mathbb{Q}} 0 = 0, \tag{10}$$

which leads to a contradiction. Therefore C is not measurable on $\mathbb{R}$. We shall also be able to CHECK THAT if $\mu(C) > 0$, then $\mu((0, 2)) = \infty$, and in general $\mu((a, b)) = \infty$, $\forall a < b$. Indeed, we have that

$$\mu((0, 2)) \geq \mu\left(\bigcup_{r\in\mathbb{Q}\cap[0,1]} C_r\right) = \sum_{r\in\mathbb{Q}\cap[0,1]} \mu(C_r) = \sum_{r\in\mathbb{Q}\cap[0,1]} \mu(C) = \infty. \tag{11}$$

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Example 1.15 (An algebra that is not a σ-algebra)**.** Let $\mathbb{N}$ be the space and take

$$\mathscr{E} = \left\{A \subseteq \mathbb{N};\ A \text{ is finite or } A^C \text{ is finite}\right\}. \tag{12}$$

If $A \in \mathscr{E}$, then either A is finite (which means $(A^C)^C$ is finite) or A^C is finite, implying that $A^C \in \mathscr{E}$. Now if $A, B \in \mathscr{E}$, we discuss two cases. If A and B are both finite, then $A \cup B$ is finite. Otherwise A^C is finite, thus $(A \cup B)^C = A^C \cap B^C$ is finite. We have shown that $\mathscr{E}$ is indeed an algebra. However, if we take $A_n = \{2n\}$, then $A = \bigcup_{n\in\mathbb{N}} A_n$ is the set of all even numbers, which is not finite and does not have a finite complement either. It follows that $\mathscr{E}$ is not a σ-algebra.

Example 1.16. Given $A_1, \cdots, A_m$ subsets of a non-empty set E, and let $\mathscr{A} := \{A_1, \cdots, A_m\}$. The goal is to describe all elements of $\mathscr{E} = \sigma(\mathscr{A})$. To do this, we denote $A_i^1 := A_i$ and $A_i^0 := A_i^C$, then for some $\epsilon \in \{0,1\}^m$, define $A_\epsilon = \bigcap_{1 \le i \le m} A_i^{\epsilon_i}$. We shall first show that

$$A \in \mathscr{E} \iff A = \bigcup_{\epsilon \in I} A_\epsilon \quad \text{for some } I \subseteq \{0,1\}^m. \tag{13}$$

$\Longleftarrow$ This is trivial by definition. Each $A_i \in \mathscr{E}$, so each $A_i^C \in \mathscr{E}$, and thus each A_ϵ, being a finite union of A_i's and their complements, is in $\mathscr{E}$. This implies that any countable union of A_ϵ's is also in $\mathscr{E}$.

$\Longrightarrow$ Let

$$\mathscr{F} = \left\{ A \subseteq E;\ A = \bigcup_{\epsilon \in I} A_\epsilon \text{ for some } I \subseteq \{0,1\}^m \right\}. \tag{14}$$

Clearly $\varnothing \in \mathscr{F}$. Take an arbitrary $A = \bigcup_{\epsilon \in I} A_\epsilon \in \mathscr{F}$, then we have that

$$A^C = \bigcup_{\epsilon \in I^C} A_\epsilon \in \mathscr{F}. \tag{15}$$

This is because all A_ϵ's are disjoint, and their union is the entire space E. Now take $A_n \in \mathscr{F}$, $n \in \mathbb{N}$, then

$$\bigcup_{n \in \mathbb{N}} A_n = \bigcup_{\epsilon \in \bigcup_{n \in \mathbb{N}} I_n} A_\epsilon \in \mathscr{F}, \tag{16}$$

thus we can conclude that $\mathscr{F}$ is a σ-algebra. Note that $\mathscr{A} \subseteq \mathscr{F}$. Indeed, if we visualize on a Venn diagram, we will see that each A_ϵ represents a single and unique area in that diagram. Each set A_i would be some union of these single areas, so $\mathscr{A} \subseteq \mathscr{F}$. Moreover, $\mathscr{E} = \sigma(\mathscr{A})$ is the smallest σ-algebra containing $\mathscr{A}$, and since $\mathscr{F}$ is an σ-algebra containing $\mathscr{A}$, we can conclude that $\mathscr{E} \subseteq \mathscr{F}$, and the proof is complete.

Now we can see that each element of $\mathscr{E}$ can be written in the form of $A = \bigcup_{\epsilon \in I} A_\epsilon$ for some $I \subseteq \{0,1\}^m$. Note that for all $\epsilon \in \{0,1\}^m$ and $A \in \mathscr{E}$, either $A \cap A_\epsilon = \varnothing$ or $A_\epsilon \subseteq A$. This is the same meaning as "a single area in the Venn diagram" as in the proof above. The subsets A_ϵ are thus called the atoms of $\mathscr{E}$.

1.1.2 Construction and Characterization of Measures

As previously mentioned, it is difficult to directly work with a σ-algebra (for instance, defining a measure). Therefore, we try to specify the values on a π-system generating it, and extend the definition to the whole σ-algebra.

Lemma 1.17. Let E be a set, $\mathscr{I}$ be a π-system on E, and $\mathscr{E} = \sigma(\mathscr{I})$. Suppose that μ_1 and μ_2 are measures on $(E, \mathscr{E})$, such that $\mu_1|_{\mathscr{I}} = \mu_2|_{\mathscr{I}}$ and $\mu_1(E) = \mu_2(E) < \infty$, then $\mu_1 = \mu_2$ on $\mathscr{E}$.

Remark 1.18. Check that if $\mathscr{I}$ is not a π-system, the uniqueness (as specified in the lemma above) can fail.

Theorem 1.19 (Carathéodory's extension theorem)**.** Let E be a set, $\mathscr{E}_0$ be an algebra on E, and $\mathscr{E} := \sigma(\mathscr{E}_0)$. Assume that $\mu_0 : \mathscr{E}_0 \to [0, \infty]$ is a countably additive set function on $\mathscr{E}_0$, then there exists a measure μ on $(E, \mathscr{E})$, such that $\mu = \mu_0$ on $\mathscr{E}_0$.

Remark 1.20. Lemma 1.17 and Theorem 1.19 are important tools for proving uniqueness and existence of measures (respectively), for instance, the Lebesgue measure.

Theorem 1.21. There exists a unique Borel measure μ on $\mathbb{R}$, such that $\forall a < b \in \mathbb{R}$, $\mu((a,b]) = b - a$. This measure μ is called the **Lebesgue measure on** $\mathbb{R}$.

1.1.3 Elementary Inequalities

Lemma 1.22. Let $(E, \mathscr{E}, \mu)$ be a measure space. Then,

$$\mu(A \cup B) \le \mu(A) + \mu(B), \qquad \mu(A \cup B) + \mu(A \cap B) = \mu(A) + \mu(B). \tag{17}$$

Proof. We have that

$$\mu(A \cup B) = \mu(A \cup (B \setminus A)) = \mu(A) + \mu(B \setminus A) \leq \mu(A) + \mu(B), \tag{18}$$
$$\mu(A \cup B) + \mu(A \cap B) = \mu(A) + \mu(B \setminus A) + \mu(A \cap B) = \mu(A) + \mu(B), \tag{19}$$

so the proof is complete. The second equation in fact implies the first inequality. □

Remark 1.23. Both observations above can be generalized to a finite number of sets. The first inequality follows by the observation that $\bigcup_{n\in\mathbb{N}} F_n = \bigcup_{n\in\mathbb{N}} \left(F_n \setminus \bigcup_{i\leq n-1} F_i\right)$. The second equality follows by induction into the inclusion-exclusion formula.

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Example 1.24. Let Ω be a non-empty set. If $\mathscr{F}_i$ is a σ-algebra for each $i \in I$, where I is a non-empty set of indices. Show that $\mathscr{F} = \bigcap_{i\in I} \mathscr{F}_i$ is a σ-algebra.

Proof. We have that $\varnothing, \Omega \in \mathscr{F}_i$, $\forall i \in I$, thus $\varnothing, \Omega \in \mathscr{F}$. Take an arbitrary $A \in \mathscr{F}$, we have that $A \in \mathscr{F}_i$, $\forall i \in I$, and thus $A^C \in \mathscr{F}_i$, $\forall i \in I$, so that $A^C \in \mathscr{F}$. Now take arbitrary $\{A_n\}_{n\geq 1} \subseteq \mathscr{F}$, then $A_n \in \mathscr{F}_i$, $\forall i \in I$, $\forall n \geq 1$. Therefore, $\bigcup_{n\geq 1} A_n \in \mathscr{F}_i$, $\forall i \in I$. This means that $\bigcup_{n\geq 1} A_n \in \mathscr{F}$, and the proof is complete. □

Example 1.25. Let Ω be a non-empty set.

(1) Show that if $\mathscr{F}_1 \subset \mathscr{F}_2 \subset \cdots$ are σ-algebras, then $\mathscr{F} = \bigcup_{n\geq 1} \mathscr{F}_n$ is an algebra.

(2) Show that $\mathscr{F}$ is not a σ-algebra.

Proof. (1) We have that $\varnothing, \Omega \in \mathscr{F}_n$, $\forall n$. Therefore, $\varnothing, \Omega \in \mathscr{F}$. Take an arbitrary $A \in \bigcup_{n\geq 1} \mathscr{F}_n$, then there exists $n_0 \geq 1$, such that $A \in \mathscr{F}_{n_0}$. This means that $A^C \in \mathscr{F}_{n_0}$, and thus $A^C \in \mathscr{F}$. Now take arbitrary $A, B \in \mathscr{F}$, then there exists $n_1, n_2 \geq 1$, such that $A \in \mathscr{F}_{n_1}$ and $B \in \mathscr{F}_{n_2}$. We have that $A \cup B \in \mathscr{F}_{\max(n_1,n_2)}$, so that $A \cup B \in \mathscr{F}$. This completes the proof.

(2) Take a counterexample $\mathscr{F}_n = \mathscr{P}(\{1, \cdots, n\})$, the set of all subsets of $\{1, \cdots, n\}$. We can see that $\mathscr{F}$ is then the set of subsets of $\mathbb{N}$ with *finite* cardinality. Now consider $A_n = \{n\} \in \mathscr{F}_n$, $n \geq 1$, then the countable union $\bigcup_{n\geq 1} A_n = \mathbb{N} \notin \mathscr{F}$. Therefore $\mathscr{F}$ is not a σ-algebra in this case. □

Example 1.26. Let μ be a finitely additive measure in an algebra $\mathscr{S}$. Consider $(A_n)_{n\geq 1}$ disjoint sets in $\mathscr{S}$ such that $A = \bigcup_{n\geq 1} A_n \in \mathscr{S}$. Show that

$$\mu(A) \geq \sum_{n=1}^{\infty} \mu(A_n). \tag{20}$$

Proof. Note that $\mathscr{S}$ is not necessarily a σ-algebra. Decompose

$$A = \bigcup_{n=1}^{\infty} A_n = \underbrace{\left(\bigcup_{n=1}^{k} A_n\right)}_{\in\mathscr{S}} \cup \underbrace{\left(\bigcup_{n=k+1}^{\infty} A_n\right)}_{\in\mathscr{S}}. \tag{21}$$

Therefore, we can see that

$$\mu(A) = \mu\left(\bigcup_{n=1}^{k} A_n\right) + \underbrace{\mu\left(\bigcup_{n=k+1}^{\infty} A_n\right)}_{\geq 0} \geq \mu\left(\bigcup_{n=1}^{k} A_n\right). \tag{22}$$

If $\mu(A) = \infty$, we are done, so we assume that $\mu(A) < \infty$. Taking $k \to \infty$, we can conclude that

$$\mu(A) \geq \mu\left(\bigcup_{n=1}^{\infty} A_n\right). \tag{23}$$

□

Example 1.27. Consider a probability space $(\Omega, \mathscr{F}, \mathbb{P})$ and a family of measurable sets $(A_n)_{n\geq 1}$ such that $\mathbb{P}(A_n) = 1$. Show that

$$\mathbb{P}\left(\bigcap_{n=1}^{\infty} A_n\right) = 1. \tag{24}$$

Proof. We can see that

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n^C\right) \leq \sum_{n=1}^{\infty} \mathbb{P}(A_n^C) = \sum_{n=1}^{\infty} 0 = 0, \tag{25}$$

so the proof is complete by taking the complement. □

Example 1.28. For a sequence of events $(A_n)_{n\geq 1}$ with $\lim_{n\to\infty} \mathbb{P}(A_n) = 1$ and $0 < c < 1$, show that there exists a subsequence $(n_k)_{k\geq 1}$ with $n_k \to \infty$, such that

$$\mathbb{P}\left(\bigcap_{k=1}^{\infty} A_{n_k}\right) > c. \tag{26}$$

Proof. Fix $\epsilon_0 = 1 - c > 0$, and take $\epsilon_i = \epsilon_0/2^i$, $\forall i \geq 1$. From the given limit, we can see that $\forall \epsilon > 0$, $\exists n_0 \geq 1$, such that $\forall n \geq n_0$, we have that $|1 - \mathbb{P}(A_n)| < \epsilon$. Therefore, we can take n_k such that $|1 - \mathbb{P}(A_{n_k})| < \epsilon_k$. Then, we have that

$$\mathbb{P}\left(\bigcup_{k=1}^{\infty} A_{n_k}^C\right) \leq \sum_{k=1}^{\infty} \mathbb{P}(A_{n_k}^C) < \sum_{k=1}^{\infty} \frac{\epsilon_0}{2^k} = \epsilon_0 = 1 - c. \tag{27}$$

Then we can conclude that

$$\mathbb{P}\left(\bigcup_{k=1}^{\infty} A_{n_k}\right) = 1 - \mathbb{P}\left(\bigcup_{k=1}^{\infty} A_{n_k}^C\right) > c \tag{28}$$

so the proof is complete. □